

FRACTAL-WAVE ALGEBRA

INVARIANTS, STABILITY AND THE GENERATION OF STRUCTURE

The recursive dynamics introduced on Page 3 does not by itself define a physical structure. A recursive process can generate unlimited configurations, including unstable, divergent or trivial states. FWA therefore requires a second condition: the existence of invariants that survive transformation while the state itself changes.

Let the recursive transformation be $\Phi_{n+1} = F(\Phi_n)$. A quantity $Q[\Phi]$ is an invariant of the transformation when

$$Q[\Phi_{n+1}] = Q[\Phi_n]$$

or, in the continuous limit,

$$dQ / dt = 0$$

The state can therefore change without destroying the structural information that defines its identity.

This distinction is fundamental. FWA does not require a stable object to remain geometrically identical. Stability means that the generating relation remains preserved through transformation.

DYNAMIC STABILITY

A stable wave structure is a configuration that remains bounded under recursive evolution.

For a state Φ_* , the condition

$$F(\Phi_*) = \Phi_*$$

defines a fixed point.

However, physical structures need not be static. A more general stable state may satisfy

$$F^k(\Phi_*) = \Phi_*$$

where $k > 1$. Such a state is periodic in the recursive domain while remaining dynamically active.

The same principle allows quasiperiodic and self-similar attractors:

$$\Phi_{n+k} \sim A \cdot \Phi_n$$

Thus stability does not imply absence of motion. It implies preservation of the structural relation during motion.

This gives the FWA distinction:

$$\textit{static equilibrium} \neq \textit{dynamic stability}$$

A physical structure may therefore exist precisely because it continuously changes while remaining within an invariant region of its state space.

THE INVARIANT CORE

Let the complete state be represented by

$$\Phi = (A, \phi, \lambda, \chi, Lu, T),$$

where A is amplitude, ϕ phase, λ scale, χ chirality or orientation, Lu the differentiation gap, and T the topological structure.

The recursive transformation becomes

$$\Phi_{n+1} = F(A_n, \phi_n, \lambda_n, \chi_n, Lu_n, T_n)$$

The individual components may change:

$$A_{n+1} \neq A_n, \quad \phi_{n+1} \neq \phi_n, \quad \lambda_{n+1} \neq \lambda_n,$$

while a higher-order invariant remains preserved:

$$I(\Phi_{n+1}) = I(\Phi_n)$$

The observable structure is therefore determined not by the absolute value of one variable but by the relationship among the variables.

This provides a natural mathematical basis for treating a particle, mode or localized excitation as a stable recursive configuration rather than as an irreducible primitive object.

FRACTAL CONSERVATION

The conservation principle can be extended across scales.

Let Q_n denote an invariant at recursion level n . FWA requires

$$Q_{n+1} = Q_n$$

for the invariant itself, while the representation of the state may transform according to

$$\Phi_{n+1} = A_n \cdot \Phi_n(\lambda_r r, \lambda_t t + \delta_n)$$

Consequently, the same invariant may appear through different geometric or temporal realizations.

This produces a hierarchy:

$$Q \rightarrow \Phi_0 \rightarrow \Phi_1 \rightarrow \Phi_2 \rightarrow \dots$$

The structure changes; the generative constraint persists.

The physical significance of this construction is that scale transformation does not necessarily create a new ontology at every level. Multiple scales can be manifestations of the same recursive rule.

INFORMATION AND INVARIANCE

If information is associated with differentiation from ideal symmetry,

$$I = \delta S,$$

then the information content of a state can be represented as a structural deviation rather than as an arbitrary numerical label.

For two successive states,

$$\Delta I = I_{n+1} - I_n$$

A dynamically conserved information structure therefore satisfies

$$\Delta I = 0$$

even when $\Phi_{n+1} \neq \Phi_n$. This is the same distinction between state change and structural identity.

A system can therefore continuously transform while preserving the information necessary to recognize the

transformation as belonging to the same recursive family.

FROM RECURSION TO PARTICLES

Within this framework, a localized physical object can be defined operationally as a stable attractor of the recursive wave dynamics.

Let $A(\Phi)$ denote the attractor set of the transformation F . A particle-like state corresponds to

$$\Phi_p \in A(\Phi)$$

Its persistence does not require an absolutely static entity. Instead,

$$\Phi_p(t + \Delta t) \neq \Phi_p(t)$$

while the state remains inside the same attractor basin:

$$\Phi_p(t) \in A$$

In this interpretation, persistence is a property of the dynamics rather than a primitive property of matter.

A particle is therefore a stable mode of recursive organization.

THE ROLE OF BOUNDARIES

A recursive structure cannot be defined only by its internal state. Its boundary determines how the state interacts with the surrounding wave configuration.

Let $B[\Phi]$ represent the boundary operator. Then

$$B[\Phi] = 0$$

can represent an idealized closed configuration, while $B[\Phi] \neq 0$ represents coupling to an external or neighboring structure.

The resulting dynamics becomes

$$\Phi_{n+1} = F(\Phi_n, B[\Phi_n])$$

The boundary therefore becomes part of the generative process.

This provides a route toward describing reflection, transmission, interference, mode conversion and chirality changes without treating them as unrelated mechanisms.

THE FWA CONDITION

The central condition of Page 4 can therefore be summarized as:

Change of state + preservation of invariant = dynamic structure

Or more formally,

$$\Phi_{n+1} = F(\Phi_n), \quad I(\Phi_{n+1}) = I(\Phi_n)$$

The first equation generates change. The second equation prevents arbitrary dissolution of structure. Together they define the minimum mathematical requirement for a persistent fractal-wave object.

Thus FWA moves from the ontology established on Pages 1–2, through the recursive dynamics of Page 3, to a criterion for why structures can persist while continuously changing.